

E-Commerce Purchase Conversion Analytics

Data Analytics with AI Internship – Capstone Project

Executive Summary

This project analyzes online shopping sessions to identify behavioral patterns associated with purchase completion. The UCI dataset contains 12,330 sessions; 1,908 ended in purchase and 10,422 did not, giving an overall conversion rate of 15.47%.

Business Problem

E-commerce session data contains useful behavioral signals, but raw activity does not directly translate into decisions. This project converts session-level behavior into conversion insights and recommended business actions.

Objectives

Measure conversion; compare visitor segments; analyze timing and engagement patterns; examine page-value signals; identify traffic opportunities; and provide an interactive dashboard.

Dataset

UCI Online Shoppers Purchasing Intention Dataset. 12,330 sessions, 18 attributes, Revenue as the binary outcome, no missing values, CC BY 4.0. DOI: 10.24432/C5F88Q.

Dashboard

The Streamlit dashboard contains KPI cards, visitor-type conversion, monthly conversion, weekday/weekend conversion, page-value conversion and filters for visitor type and month.

Business Actions

Monitor conversion as the primary KPI; segment new and returning visitors; use behavioral intent signals for targeted conversion experiences; use timing patterns for campaigns; and review low-conversion traffic sources.

Limitations

The dataset does not include profit, customer lifetime value or advertising cost. The analysis therefore focuses on conversion behavior rather than profitability.

Conclusion

The project implements a reproducible Data → Information → Insight → Decision → Action workflow and can be extended with predictive modeling or additional business data.

KPI Snapshot

KPI	Value
Total sessions	12,330
Purchase sessions	1,908
Non-purchase sessions	10,422
Overall conversion rate	15.47%

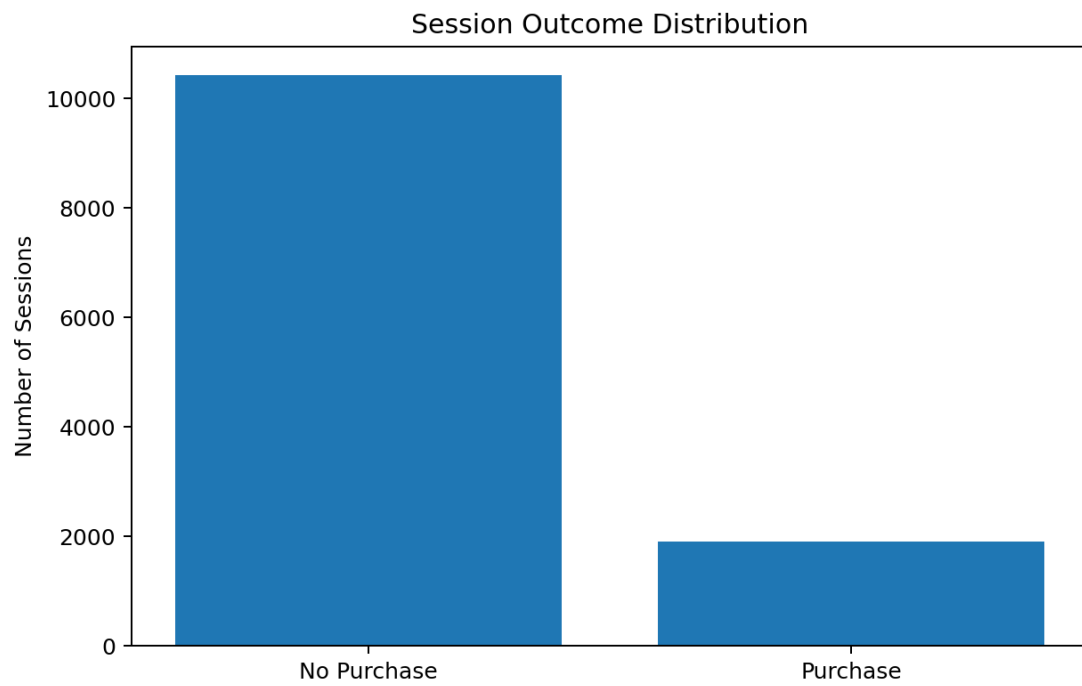


Figure 1. Purchase vs non-purchase session distribution.

Reproducibility note: run `analysis.py` once to generate the full KPI and conversion tables under `outputs/`, then run `app.py` to open the interactive dashboard. Capture the final dashboard for the submission report.